

A lower bound for stepsize-based acceleration of gradient descent

Jianhao Ma

Tsinghua University

Joint work with Yuxin Chen (University of Pennsylvania)

Center for Statistical Science, Peking University

September 24, 2026

GD for smooth convex functions

Assume $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and L -smooth, with a minimizer $x_\star$.

Gradient descent (GD) uses the update

$$x_{t+1} = x_t - \eta_t \nabla f(x_t)$$

GD for smooth convex functions

Assume $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and L -smooth, with a minimizer $x_\star$.

Gradient descent (GD) uses the update

$$x_{t+1} = x_t - \eta_t \nabla f(x_t)$$

Textbook result (Bubeck, 2015)

For the constant stepsize $\eta_t = 1/L$, the last iterate satisfies

$$f(x_T) - f(x_\star) \leq \frac{2L \|x_0 - x_\star\|^2}{T} = O(T^{-1}).$$

Nesterov's GD

Starting from $x_0 = y_0$, Nesterov's method uses

$$\begin{aligned}x_t &= y_{t-1} - \frac{1}{L} \nabla f(y_{t-1}), \\y_t &= x_t + \frac{t-1}{t+2} (x_t - x_{t-1})\end{aligned}$$

Nesterov's GD

Starting from $x_0 = y_0$, Nesterov's method uses

$$\begin{aligned}x_t &= y_{t-1} - \frac{1}{L} \nabla f(y_{t-1}), \\y_t &= x_t + \frac{t-1}{t+2} (x_t - x_{t-1})\end{aligned}$$

Theorem (Nesterov, 1983)

After T iterations, the last iterate of Nesterov's GD satisfies

$$f(x_T) - f(x_\star) \leq \frac{2L \|x_0 - x_\star\|^2}{(T+1)^2} = O(T^{-2}).$$

Nesterov's GD

Theorem (Nesterov, 1983)

After T iterations, the last iterate of Nesterov's GD satisfies

$$f(x_T) - f(x_*) \leq \frac{2L \|x_0 - x_*\|^2}{(T+1)^2} = O(T^{-2}).$$

- The $O(T^{-2})$ rate is optimal in the first-order oracle model.

Nesterov's GD

Theorem (Nesterov, 1983)

After T iterations, the last iterate of Nesterov's GD satisfies

$$f(x_T) - f(x_*) \leq \frac{2L \|x_0 - x_*\|^2}{(T+1)^2} = O(T^{-2}).$$

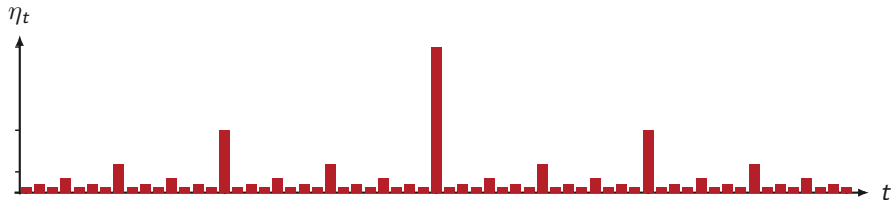
- The $O(T^{-2})$ rate is optimal in the first-order oracle model.
- Can plain GD attain this accelerated rate using stepsizes alone?

GD with silver stepsize

Result (Altschuler–Parrilo, 2025; Grimmer–Shu–Wang, 2025)

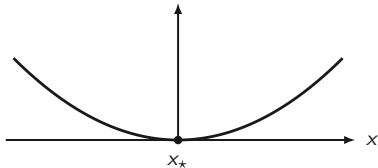
For every convex L -smooth objective, plain GD with the silver stepsize schedule achieves

$$f(x_T) - f(x_\star) = O\left(\frac{L \|x_0 - x_\star\|^2}{T^{\log_2(1+\sqrt{2})}}\right) = O\left(\frac{L \|x_0 - x_\star\|^2}{T^{1.2716\dots}}\right).$$

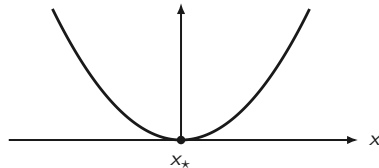


Main idea: stepsize hedging

Flat direction

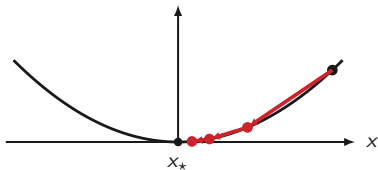


Steep direction

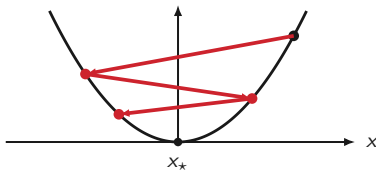


Main idea: stepsize hedging

Flat direction



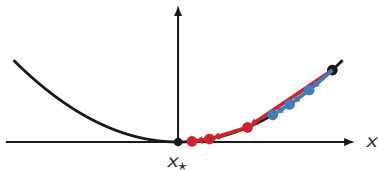
Steep direction



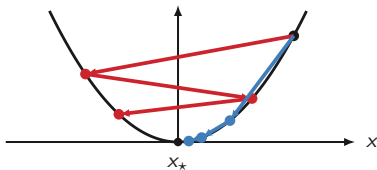
- Large stepsizes converge faster along low-curvature (flat) directions.

Main idea: stepsize hedging

Flat direction



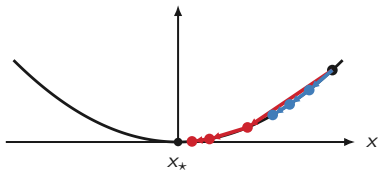
Steep direction



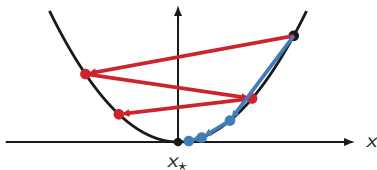
- Large stepsizes converge faster along low-curvature (flat) directions.
- Small stepsizes converge faster along high-curvature (steep) directions.

Main idea: stepsize hedging

Flat direction

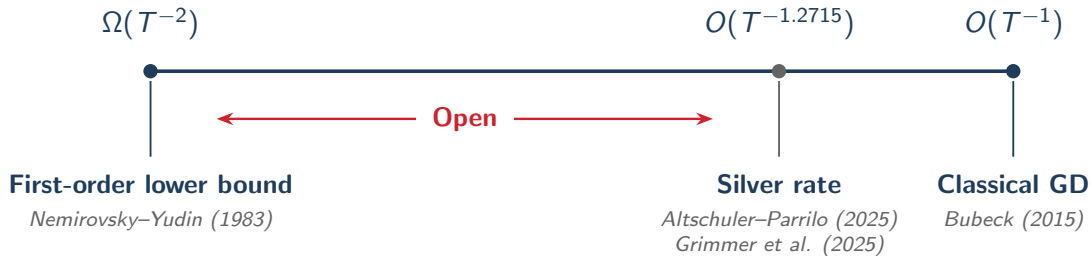


Steep direction

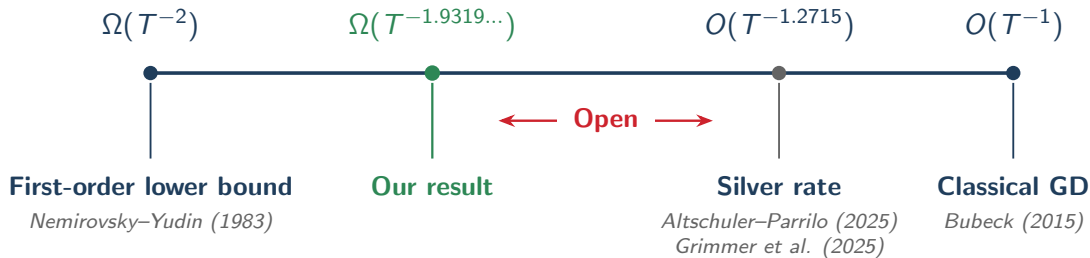


- Large stepsizes converge faster along low-curvature (flat) directions.
- Small stepsizes converge faster along high-curvature (steep) directions.
- Hedging both long and short stepsizes leads to acceleration.

Open problem

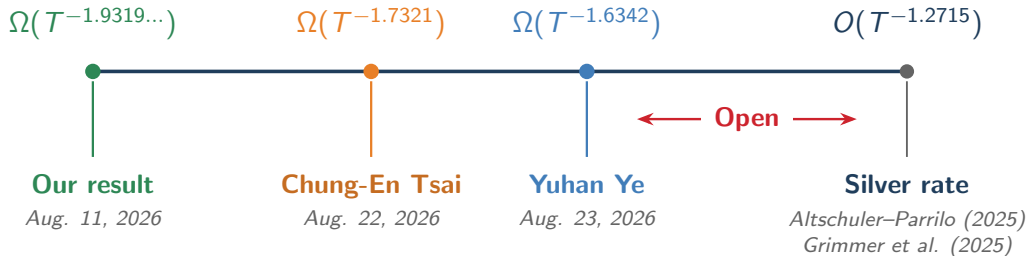


Our result



Proof found by **GPT-5.6 Sol Pro**; verified by the authors; formalized in **Lean**.

Progress in August 2026



The silver gap is closed



The silver gap is closed

- The optimal **polynomial exponent** is now settled, up to subpolynomial factors.
- Both results use **different hard instances** from ours.

Beyond GD: Heavy Ball

Heavy Ball (Polyak, 1964) adds the previous displacement:

$$x_{t+1} = x_t - \eta_t \nabla f(x_t) + \underbrace{\beta_t (x_t - x_{t-1})}_{\text{momentum}}, \quad x_{-1} = x_0.$$

- **Convex quadratics:** suitable time-varying parameters achieve $O(T^{-2})$.
Flammarion–Bach (2015); Orvieto (2025)
- **General smooth convex functions:** classical guarantees of order $O(T^{-1})$; silver GD already improves this to $O(T^{-1.2715})$.

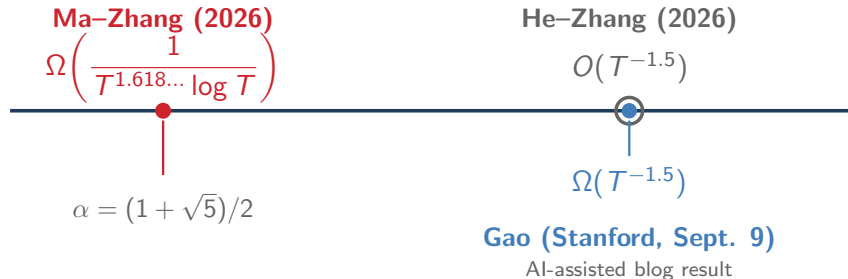
The missing lower bound (before September 2026)

The quadratic $\Omega(T^{-2})$ bound cannot separate Heavy Ball from Nesterov.

For **arbitrary predetermined schedules**, a separation on **general smooth convex functions** was open.

Heavy Ball: rate landscape

Last-iterate objective gap on smooth convex functions



Matching $T^{-3/2}$ exponents

Lower bounds: deterministic schedules, $\eta_t \geq 0$, $0 \leq \beta_t < 1$.

Upper bound: randomized schedules, in expectation; β_t may exceed 1.

Formal result

For a fixed stepsize schedule $\eta = (\eta_0, \dots, \eta_{T-1})$, define

$$R_T(\eta) := \sup_{d \in \mathbb{N}_+} \sup_{f \in \mathcal{F}_{0,L}(\mathbb{R}^d)} \sup_{x^* \in \operatorname{argmin} f} \sup_{\substack{x_0 \in \mathbb{R}^d \\ x_0 \neq x^*}} \frac{f(x_T) - f(x^*)}{\frac{1}{2}L \|x_0 - x^*\|^2}$$

- $\mathcal{F}_{0,L}(\mathbb{R}^d)$ is the class of convex, L -smooth objectives with a minimizer;
- x_T is the last GD iterate generated by η from x_0 .

Formal result

Theorem 1 (Ma–Chen, 2026)

For every $p \in (1.9319\dots, 2)$, there exists $c_p > 0$ such that, for every **predetermined nonnegative stepsize schedule** η , we have

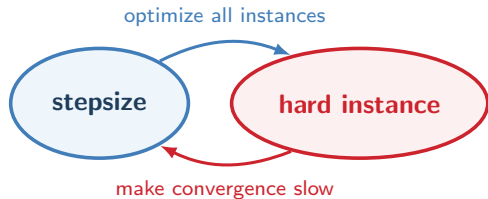
$$R_T(\eta) \geq \frac{c_p}{T^p}.$$

- the hard instance has dimension $d \leq T + 1$;
- the guarantee concerns the last iterate x_T ;
- the stepsize schedule is predetermined and nonnegative;
- the theorem covers $p > 1.9319\dots$, not the endpoint.

Talk roadmap

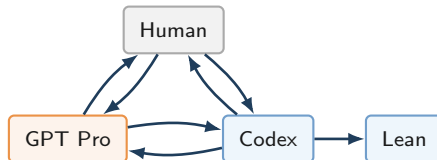
PART I

The hard instance



PART II

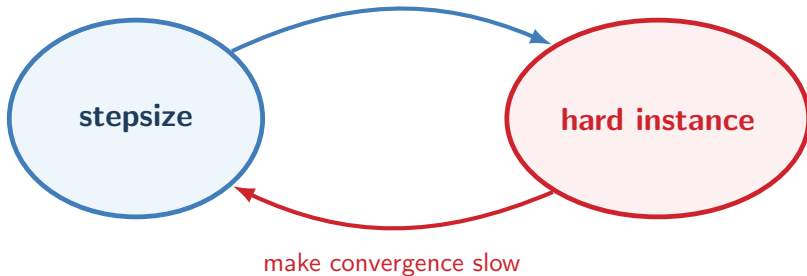
Human-AI workflow



Part I

The hard instance

optimize all the hard instances



First candidate: quadratics

$$f(x) := \frac{1}{2} \left[x_1^2 + \sum_{i=1}^{d-1} (x_i - x_{i+1})^2 + x_d^2 \right] - x_1$$

Lower bound for span-respecting methods

For every span-respecting method and $T \geq 1$, there is $d \geq T + 1$ such that, for every prescribed $x_0 \in \mathbb{R}^d$, there exist $f \in \mathcal{F}_{0,L}(\mathbb{R}^d)$ and $x_\star \in \arg \min f$ with

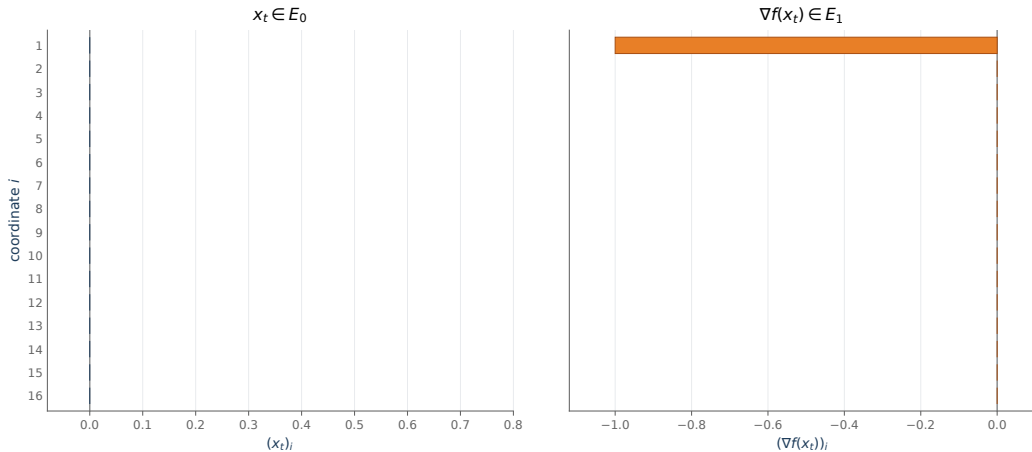
$$f(x_T) - f(x_\star) \geq c \frac{L \|x_0 - x_\star\|^2}{(T + 1)^2}$$

for a universal constant $c > 0$.

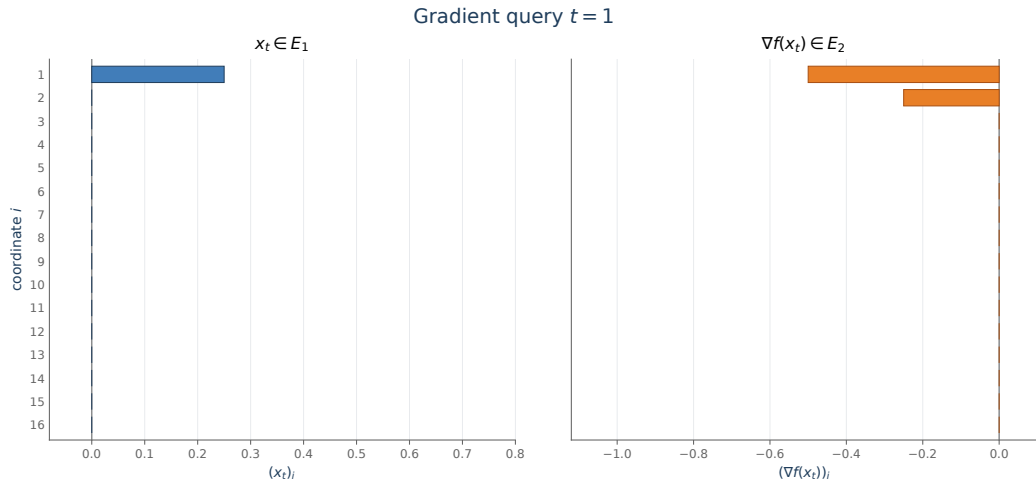
$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}$$

One new coordinate per query

Gradient query $t = 0$

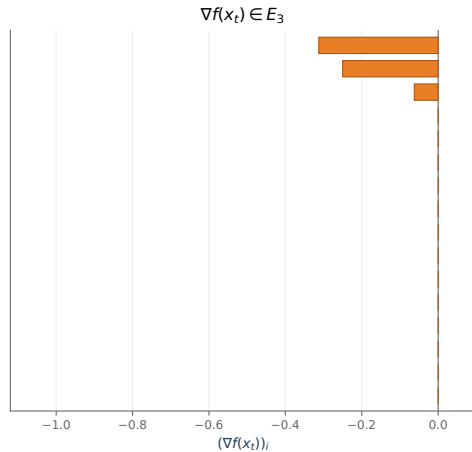
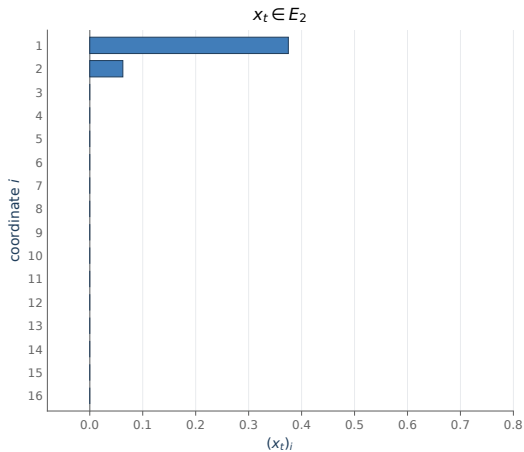


One new coordinate per query



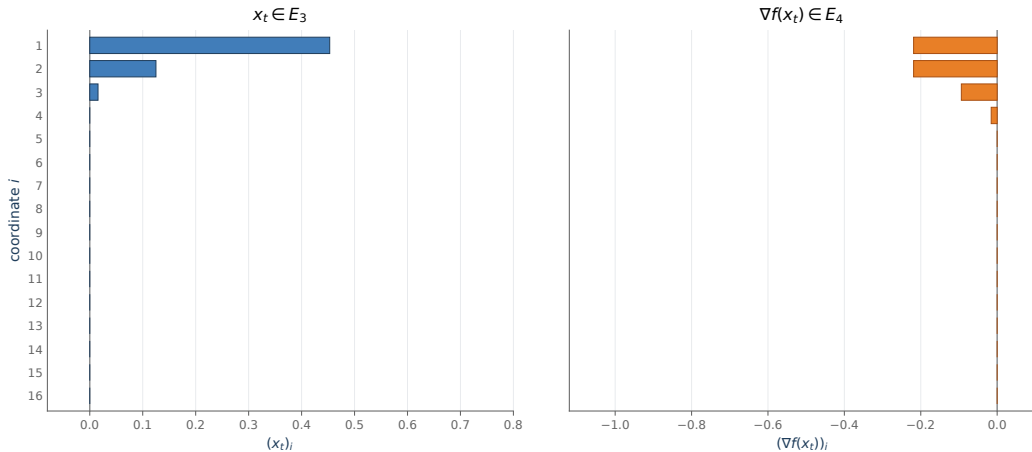
One new coordinate per query

Gradient query $t = 2$



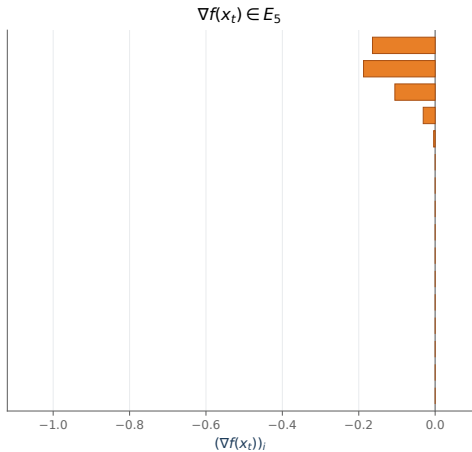
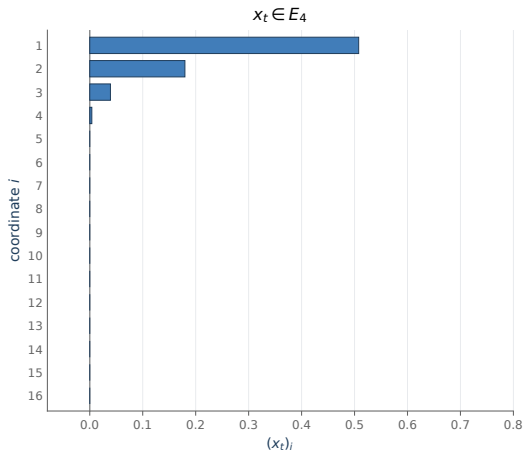
One new coordinate per query

Gradient query $t = 3$

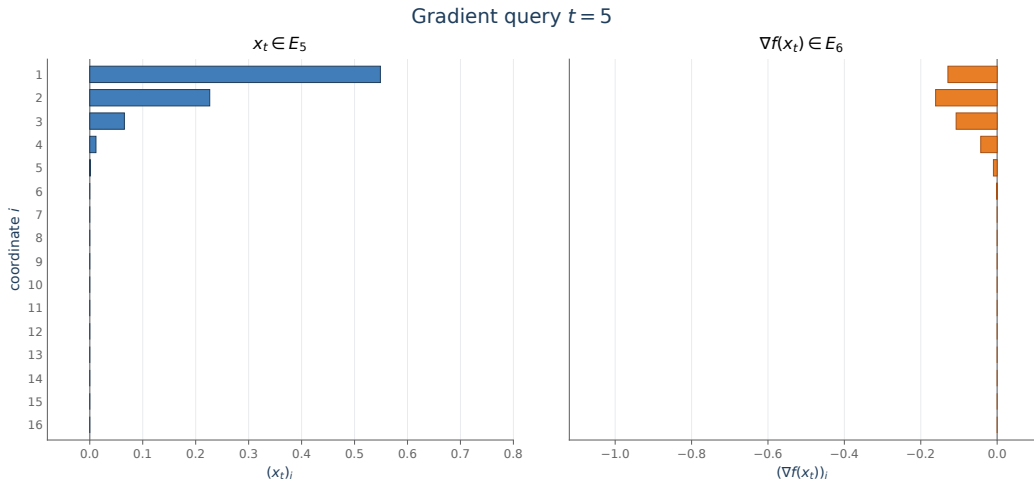


One new coordinate per query

Gradient query $t = 4$

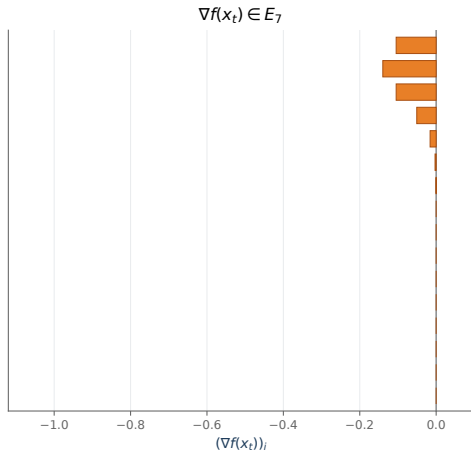
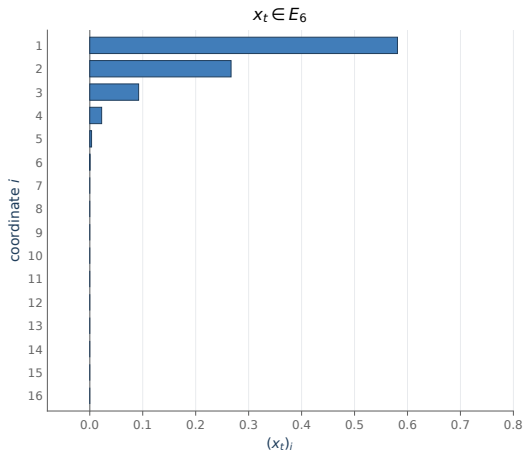


One new coordinate per query



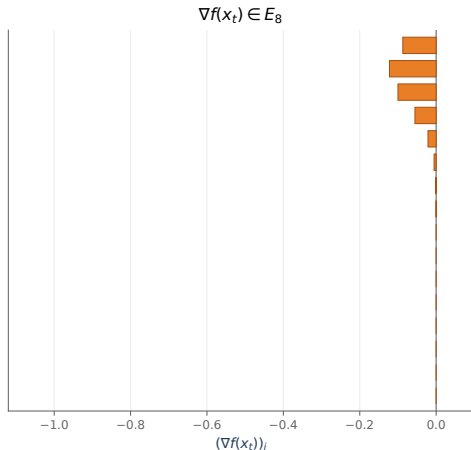
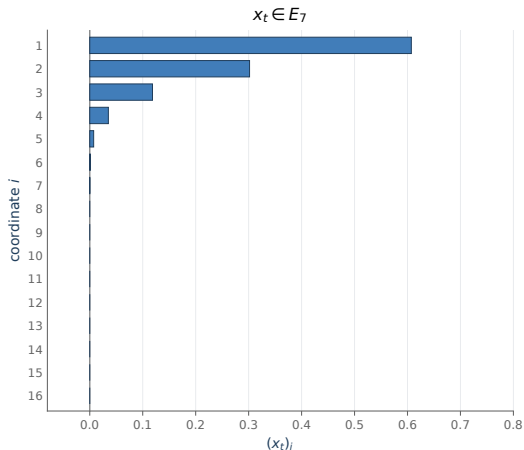
One new coordinate per query

Gradient query $t = 6$

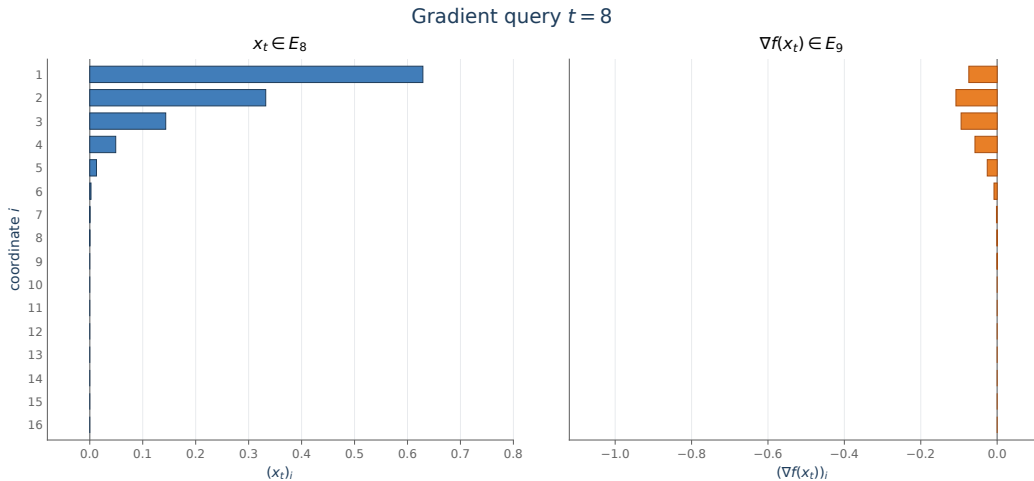


One new coordinate per query

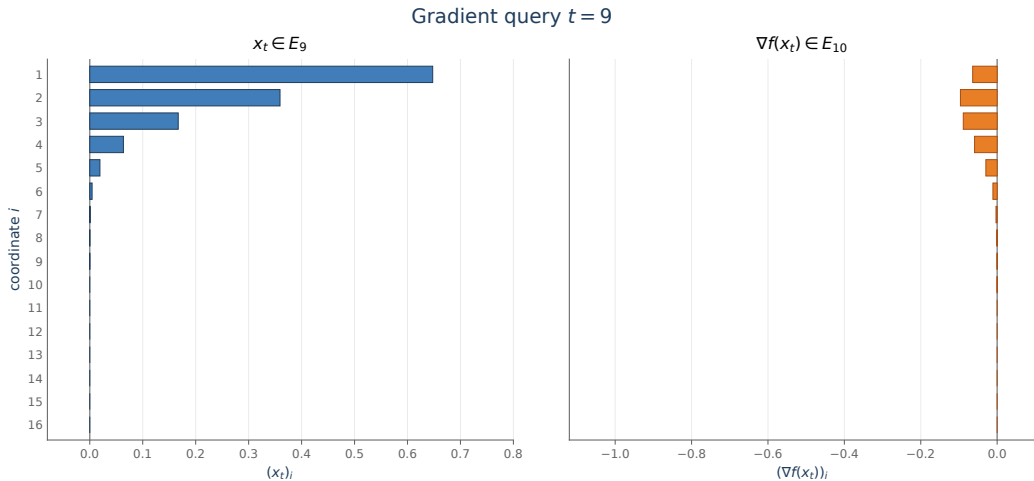
Gradient query $t = 7$



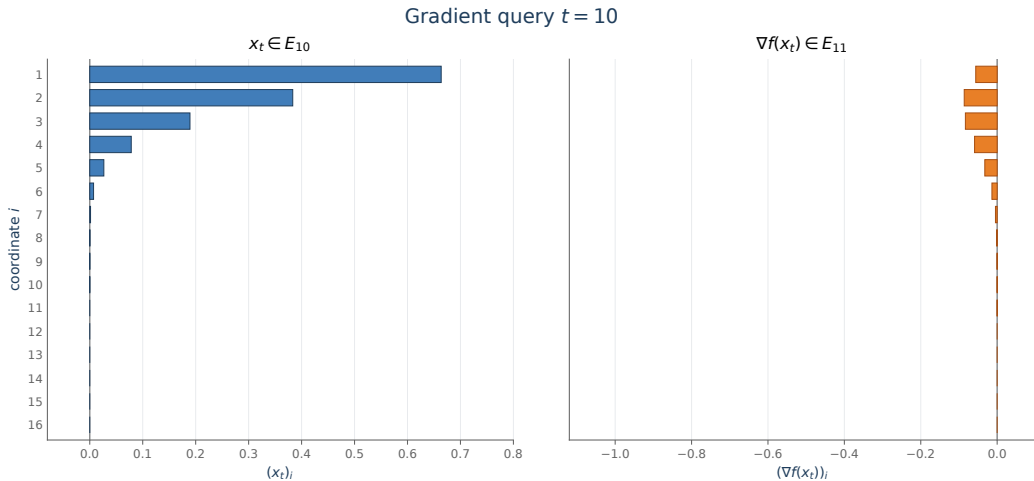
One new coordinate per query



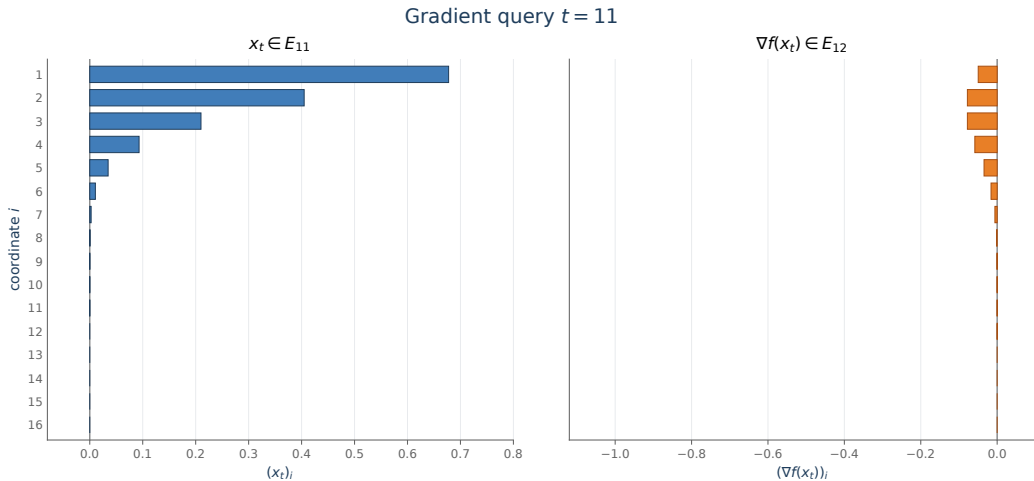
One new coordinate per query



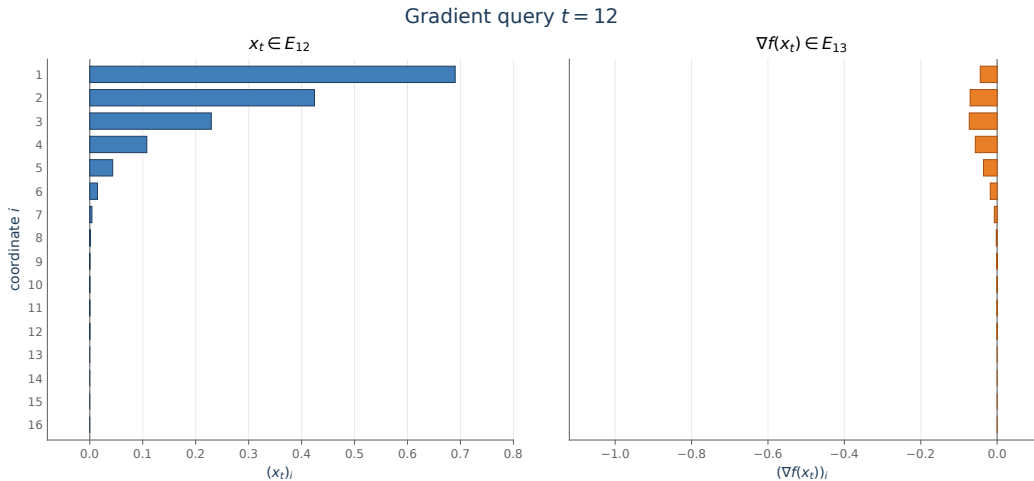
One new coordinate per query



One new coordinate per query



One new coordinate per query



Quadratics are simple

Theorem (Chebyshev acceleration)

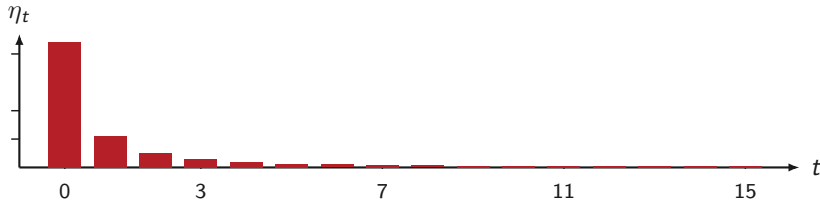
Nemirovsky (1992)

Let $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ with $0 \preceq A \preceq LI$ and minimizer x_* . For any prescribed horizon T , plain GD with

$$\eta_t = \frac{1}{L \sin^2 \left(\frac{(t+1)\pi}{2T+1} \right)}, \quad t = 0, \dots, T-1,$$

satisfies

$$f(x_T) - f^* \leq \frac{L \|x_0 - x_*\|^2}{2(2T+1)^2} = O(T^{-2}).$$



Larger stepsizes accelerate convergence

Theorem (bounded-step quadratic obstruction)

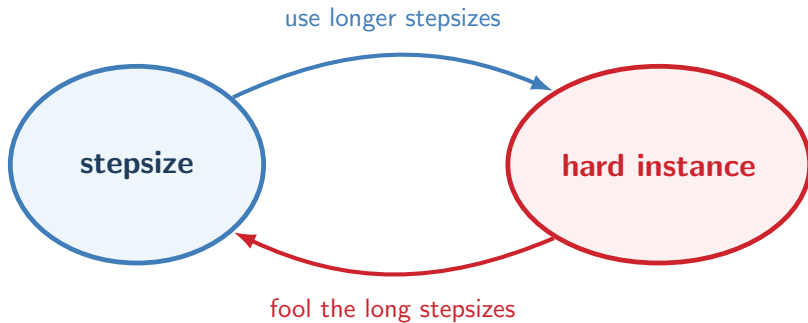
Tsai et al. (2026, Lemmas 2.1, 2.3)

For fixed $C > 0$, $\alpha \geq 0$, and any stepsize schedule satisfying $\max_{1 \leq t \leq T} \eta_t \leq CT^\alpha/L$, the worst-case last-iterate error over convex L -smooth quadratics is

$$f(x_T) - f^* = \Omega\left(\frac{L \|x_0 - x_\star\|^2}{T^{\min\{1+\alpha/2, 2\}}}\right).$$

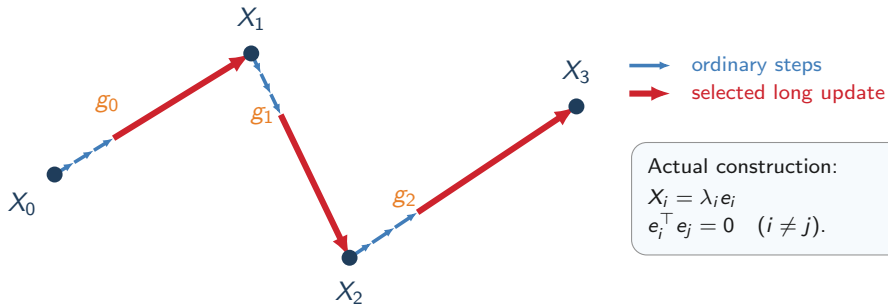
- We need to find a **non-quadratic hard instance**.
- We only need to consider **long stepsizes**.

Resisting oracle approach



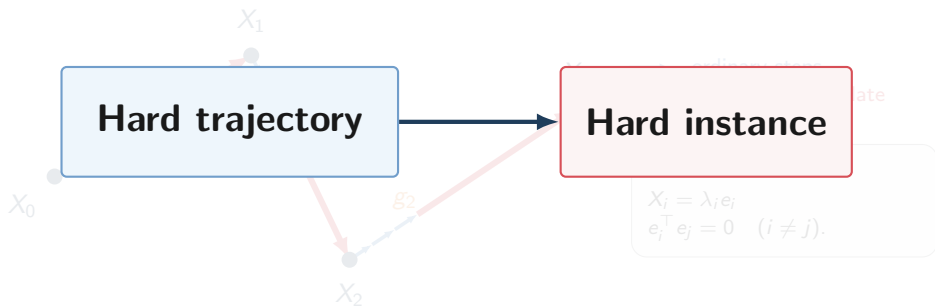
The hard trajectory

We try to **hide as much information as possible** from the (selected) long stepsizes.



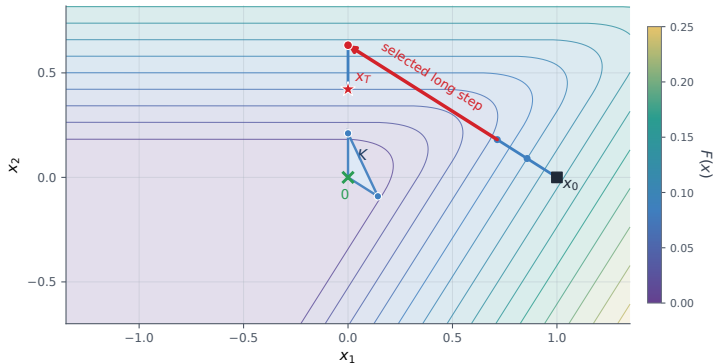
The hard trajectory

We try to **hide as much information as possible** from the (selected) long stepsizes.



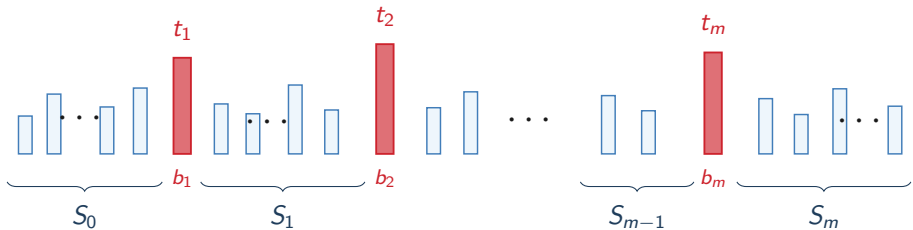
The hard function

$$F(x) = \frac{1}{2} \|x\|^2 - \frac{1}{2} \text{dist}(x, K)^2, \quad K = \text{conv}\{0, g_0, \dots, g_m\}, \quad \nabla F(x) = \Pi_K(x).$$



Two-dimensional Ma–Chen construction for $h = (1, 1, 5, 1)$; $\gamma^2 = 2/5$.

Payoff from selected long steps

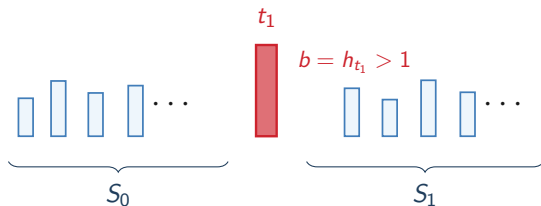


Theorem 4.1 (Ma–Chen, 2026)

$$R_T(\eta) \geq \frac{1}{S_{m-1} + b_m + 2S_m + 1} \prod_{i=1}^{m-1} \frac{S_i + b_{i+1}}{S_{i-1} + b_i + S_i + b_{i+1}} \prod_{i=1}^m \frac{b_i - 1}{S_{i-1} + 1}.$$

Long-step excess helps.

Why select a long step?



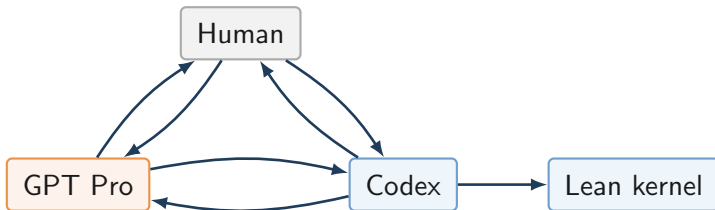
Theorem 4.1 with one selected step

$$R_T(\eta) \geq \frac{1}{S_0 + 1} \frac{b - 1}{b + C}, \quad C := S_0 + 2S_1 + 1.$$

A larger selected step gives a stronger lower bound.

Part II

Human-AI workflow



Fun facts

- Initially, we tried to prove the lower bound for strongly convex functions, but failed.

Fun facts

- Initially, we tried to prove the lower bound for strongly convex functions, but failed.
- We used GPT-5.5 Pro but did not succeed. Soon after GPT-5.6 was released, we obtained an initial proof.

Fun facts

- Initially, we tried to prove the lower bound for strongly convex functions, but failed.
- We used GPT-5.5 Pro but did not succeed. Soon after GPT-5.6 was released, we obtained an initial proof.
- The proof was not derived in a single attempt; it emerged through repeated attempts and revisions.

Fun facts

- Initially, we tried to prove the lower bound for strongly convex functions, but failed.
- We used GPT-5.5 Pro but did not succeed. Soon after GPT-5.6 was released, we obtained an initial proof.
- The proof was not derived in a single attempt; it emerged through repeated attempts and revisions.
- Along the way, we proposed several proof strategies, but ultimately none were adopted.

Fun facts

- Initially, we tried to prove the lower bound for strongly convex functions, but failed.
- We used GPT-5.5 Pro but did not succeed. Soon after GPT-5.6 was released, we obtained an initial proof.
- The proof was not derived in a single attempt; it emerged through repeated attempts and revisions.
- Along the way, we proposed several proof strategies, but ultimately none were adopted.
- Understanding the proof was difficult: it took longer than finding it.

Human effort

1. **Prompting:** We supplied the problem statement and the general proof idea.
2. **Pruning:** During discussions with ChatGPT, we acted as a pruner and terminated unpromising approaches early.
3. **Proofreading:** We proofread extensively and invested substantial effort in making the proof readable.

A principled workflow: Danus as an example

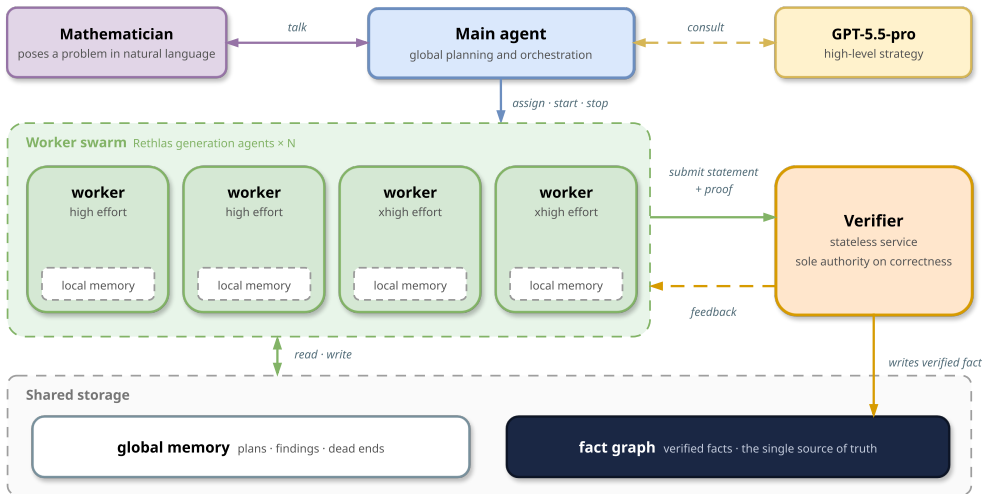


Figure 1 from Liu et al. (2026).

Takeaways

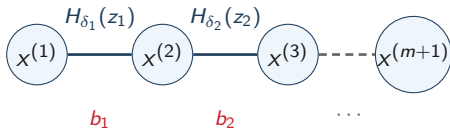
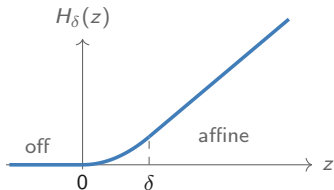
1. **An interesting result:** Stepsizes alone cannot accelerate plain GD all the way to the optimal $O(1/T^2)$ rate.
2. **A powerful construction:** It extends beyond our setting—even to anytime GD; see the blogs by [Tsai \(2026\)](#) and [Ye \(2026\)](#).
3. **A powerful proof tool:** LLMs can accelerate proof search, exposition, and formal verification.

Paper: [arXiv:2608.10418](#)

Lean: [github.com/jianhaoma/gd-lower-bound-lean](#)

Backup: a Huber chain

Each selected checkpoint activates the next link in the chain.



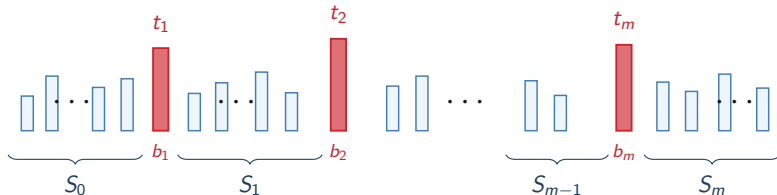
In each gap one term is affine; a checkpoint awakens the next.

Jung–Cho–Yun (2026), Eq. (4)

$$F(x) = \frac{1}{4} \sum_{i=1}^m H_{\delta_i} \left(x^{(i)} - x^{(i+1)} - c_i \right) + \frac{1}{2} H_{\delta_{m+1}} \left(x^{(m+1)} - c_{m+1} \right).$$

$H_\delta(z) = 0$ for $z \leq 0$; $z^2/2$ for $0 \leq z \leq \delta$; $\delta z - \delta^2/2$ for $z \geq \delta$. The constants c_i, δ_i depend on the schedule.

Backup: checkpoint payoff for the Huber chain



Lemma 2.3 (Jung–Cho–Yun, 2026)

$$R_T(\eta) \geq \frac{1}{4(1 + S_m)} \prod_{i=1}^m \underbrace{\left(\frac{b_i}{2(2 + S_{i-1})} \right)^2}_{\text{checkpoint transfer}}.$$